

Course Structure and Content: Biostatistics (summer term 2026)

The *Biostatistics* course will begin with foundational topics, including study design and the logic of statistical hypothesis testing. Building on these fundamentals, we will transition to practical applications using RStudio. Students will work with a synthetic dataset (*biostatistics_dataset.xlsx*) to address typical research questions, such as:

- Which patient groups appear to be more affected by certain health issues?
- Which drug seems to be the most effective?
- Is smoking associated with worse health outcomes?
- Do men have higher blood pressure than women?
- Can we predict patient recovery time?

These questions will be explored using a range of statistical methods. A key objective of the course is to help students develop routine and confidence in applied data analysis within a biomedical context. Practical work will be supported by theoretical instruction as well as supplementary online materials to reinforce understanding.

Towards the end of the course, students will consolidate and present their knowledge in a group project. Assessment will be based on two components:

1. Continuous documentation of learning progress, including methods introduced and applied from the first session onward.
2. A final group presentation demonstrating the application of biostatistical concepts.

For questions or feedback, please contact andreas.widenhorn@th-deg.de